Chapter1: Introduction to Electrical Power Systems

Topics to be covered:

Generation: Conventional and Non-conventional energy sources, generation of power from Hydro, thermal (coal based), nuclear, solar and wind (Block diagram approach), Distributed generation and its benefits.

Transmission and Distribution: Primary & Secondary transmission and distribution voltage levels with Single line diagram, concept of grid and the need for interconnection of grids. Conditions for grid Connection, Smart grid- Definition and benefits.

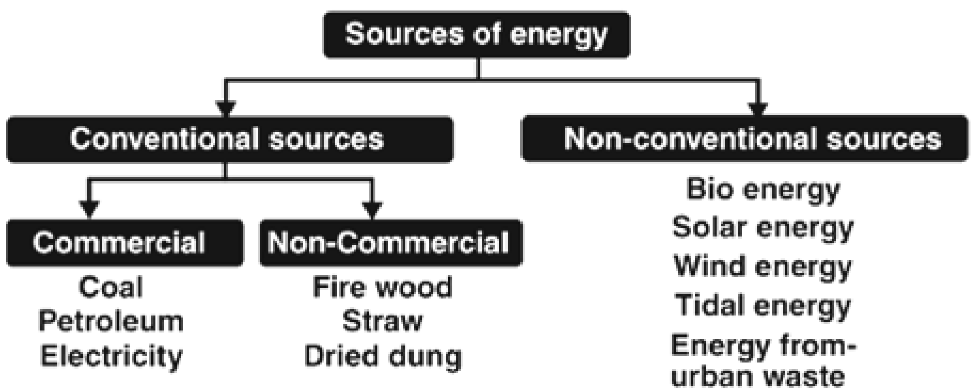
Regulatory bodies- Central and State in electrical power sector and their roles. Power tariff- Definition and types.

Introduction: Energy resources are all forms of fuels used in the modern world, either for heating, generation of electrical energy, or for other forms of energy conversion processes. Energy is one of the major parts of the economic infrastructure, being the basic input needed to sustain the economic growth. Human civilization relies on different sources of energy.

The two major sources of energy can be classified under:

1. Conventional Sources

2. Non-Conventional Sources



Conventional Sources of Energy

These sources of energy are also known as non-renewable sources of energy and are available in limited quantity. Further it can be classified as commercial and non-commercial energy.

Commercial Energy Sources

The coal, electricity and petroleum are known as commercial energy since the consumer needs to pay its price to buy them.

a) Coal: Coal is the most important source of energy. India is the fourth largest coal-producing country and the deposits are mostly found in Bihar, Orissa, Madhya Pradesh and Bengal.

b) Oil and Natural Gas: Today oil is considered to be the liquid gold and one of the crucial sources of energy in India and the world. Oil is mostly used in planes, automobiles, trains and ships. It is mainly found in Assam, Gujarat and Mumbai.

c) Electricity: Electricity is a common source of energy and used for domestic and commercial purposes. The major sources of power generation are Nuclear, Thermal Power, Hydro-electric etc.

Non-Conventional Sources of Energy

These non-conventional sources are also known as renewable sources of energy. The examples include solar energy, bioenergy, tidal energy and wind energy.

1. Solar Energy: This is the energy that is produced by the sunlight. The energy is utilized for cooking, distillation of water and to produce electricity.

2. Wind Energy: Wind energy is generated by harnessing the power of wind and mostly used in operating water pumps for irrigation purposes. India stands the second largest country in the generation of wind power.

3. Tidal Energy: The energy that is generated by exploiting the tidal waves of the sea is known as tidal energy. This source is yet to be tapped due to the lack of cost-effective technology.

4. Bio Energy: This type of energy is obtained from organic matter. It is of two kind:

(i) Bio Gas: Bio Gas is obtained from Gobar Gas Plant by putting cow dung into the plant. Besides producing gas this plant converts gobar into manure. It can be used for cooking, lighting and generation of electricity.

(ii) Bio Mass: It is also of a source of producing energy through plants and trees. The purpose of bio mass programme is to encourage afforestation for energy.

5. Energy from Urban Waste: Urban waste poses a big problem for its disposal. Now it can be used for generation of power.

Differences between Conventional and Non-Conventional Sources of Energy

1. Conventional sources of energy, as the name suggests, are those sources which are widely used all around the world since ages. On the contrary, non-conventional sources of energy are described as the energy sources whose evolution has been done in the recent past and has gained popularity since then.
2. As the conventional sources of energy are limited in nature, and their formation takes millions of years, they can be exhausted one day. Conversely, non-conventional sources of energy are the sources that are in abundance in the environment and are easily renewable, so they are inexhaustible.
3. Conventional sources of energy pollute the environment on a large scale through the smoke and hazardous waste emitted from the power plants. However, the energy produced from running water does not pollute the environment. On the other hand, non-conventional sources of energy are environment-friendly, so they do not harm the nature, by polluting it.
4. The energy produced from conventional sources are highly used for industrial and commercial purposes. As against, the energy generated out of non-conventional sources are used for domestic purposes.
5. Conventional sources of energy are costly because they are scarce but their uses are unlimited. In contrast, non-conventional sources of energy are less expensive, because of their enormous presence in nature.

Electric Power generation

Electric power generation is the process of generating electric power from sources of primary energy. Electricity is not freely available in nature, so it must be "produced" (that is, transforming other forms of energy to electricity). Production is carried out in power stations (also called "power plants"). Electricity is most often generated at a power plant by electromechanical generators, primarily driven by heat engines fueled by combustion or nuclear fission but also by other means such as the kinetic energy of flowing water and wind. Other energy sources include solar photovoltaics and geothermal power.

Hydroelectric Power Plant:

Generation of electricity by hydropower (potential energy in stored water) is one of the cleanest methods of producing electric power. Hydroelectricity is the most widely used form of renewable energy. It is a flexible source of electricity and also the cost of electricity generation is relatively low. Figure A shows the block diagram of a hydroelectric power plant and figure B shows the typical layout of the hydroelectric power plant and its basic components.

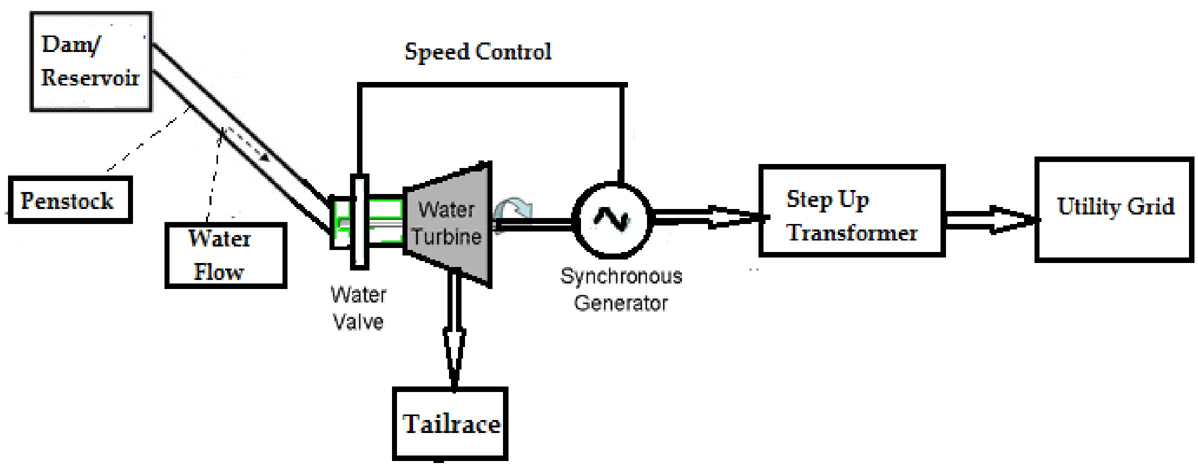


Figure A: Block diagram of a hydroelectric power

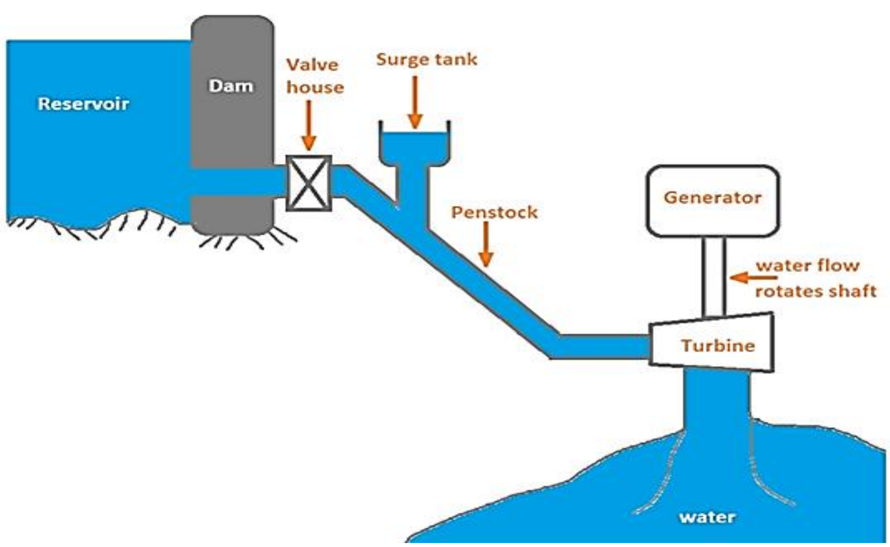


Figure B: Typical layout of the hydroelectric power plant

Dam and Reservoir: The dam is constructed on a large river in hilly areas to ensure sufficient water storage at height. The dam forms a large reservoir. The height of water level (called as water head) in the reservoir determines how much of potential energy is stored in it.

Control Gate: Water from the reservoir is allowed to flow through the penstock to the turbine. The amount of water which is to be released in the penstock can be controlled by a control gate. When the control gate is fully opened, maximum amount of water is released through the penstock.

Penstock: A penstock is a huge steel pipe which carries water from the reservoir to the turbine. Potential energy of the water is converted into kinetic energy as it flows down through the penstock due to gravity.

Surge Tank: Surge tanks are usually provided in high or medium head power plants when considerably long penstock is required. A surge tank is a small reservoir or tank which is open at the top. It is fitted between the reservoir and the power house. The water level in the surge tank rises or falls to reduce the pressure swings in the penstock. When there is sudden reduction in load on the turbine, the governor closes the gates of the turbine to reduce the water flow. This causes pressure to increase abnormally in the penstock. This is prevented by using a surge tank, in which the water level rises to reduce the pressure. On the other hand, the surge tank provides excess water needed when the gates are suddenly opened to meet the increased load demand.

Power House: The powerhouse accommodates the water turbine, generator, transformer, and control room.

Water Turbine: Water from the penstock is taken into the water turbine. The turbine is mechanically coupled to an electric generator. Kinetic energy of the water drives the turbine. There are two main types of water turbine; (i) Impulse turbine and (ii) Reaction turbine. Impulse turbines are used for large heads and reaction turbines are used for low and medium heads.

Tailrace: Tailrace is a water path to lead the water discharged from the turbine to the river or canal. The water held in the tailrace is called the Tailrace water level.

Generator: A generator is mounted in the power house and it is mechanically coupled to the turbine shaft. When the turbine blades are rotated, it drives the generator and electricity is generated which is then stepped up with the help of a transformer for the transmission purpose.

Working:

Hydroelectric power plant (Hydel plant) utilizes the potential energy of water stored in a dam built across the river. The potential energy of the stored water is converted into kinetic energy by first passing it through the penstock pipe. The kinetic energy of the water is then converted into mechanical energy in a water turbine. The turbine is coupled to the electric generator. The mechanical energy available at the shaft of the turbine is converted into electrical energy by means of the generator.

Thermal Power Plant (Coal Based):

Those power stations which convert chemical energy of fuel (coal, gas etc.) into electrical energy are called thermal power stations. The fuel used in thermal power stations is coal or gas. The heat of combustion of coal is utilized to convert water into steam which runs the steam turbine coupled with the alternator produces electrical energy. Thus, a thermal power station may sometimes called as a steam power station. Almost two third of electricity requirement of the world is fulfilled by thermal power plants (or thermal power stations). Figure A shows the block diagram of a Thermal power plant and figure B shows the typical layout of the Thermal power plant and its basic components.

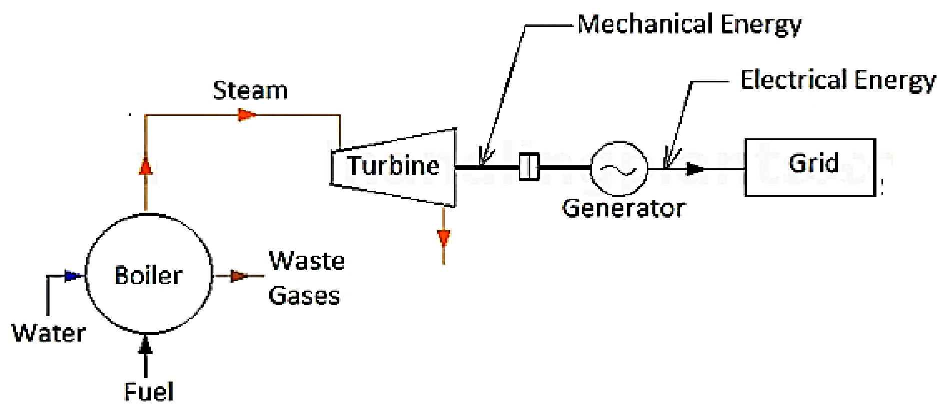


Figure A: Block diagram of a Thermal power plant

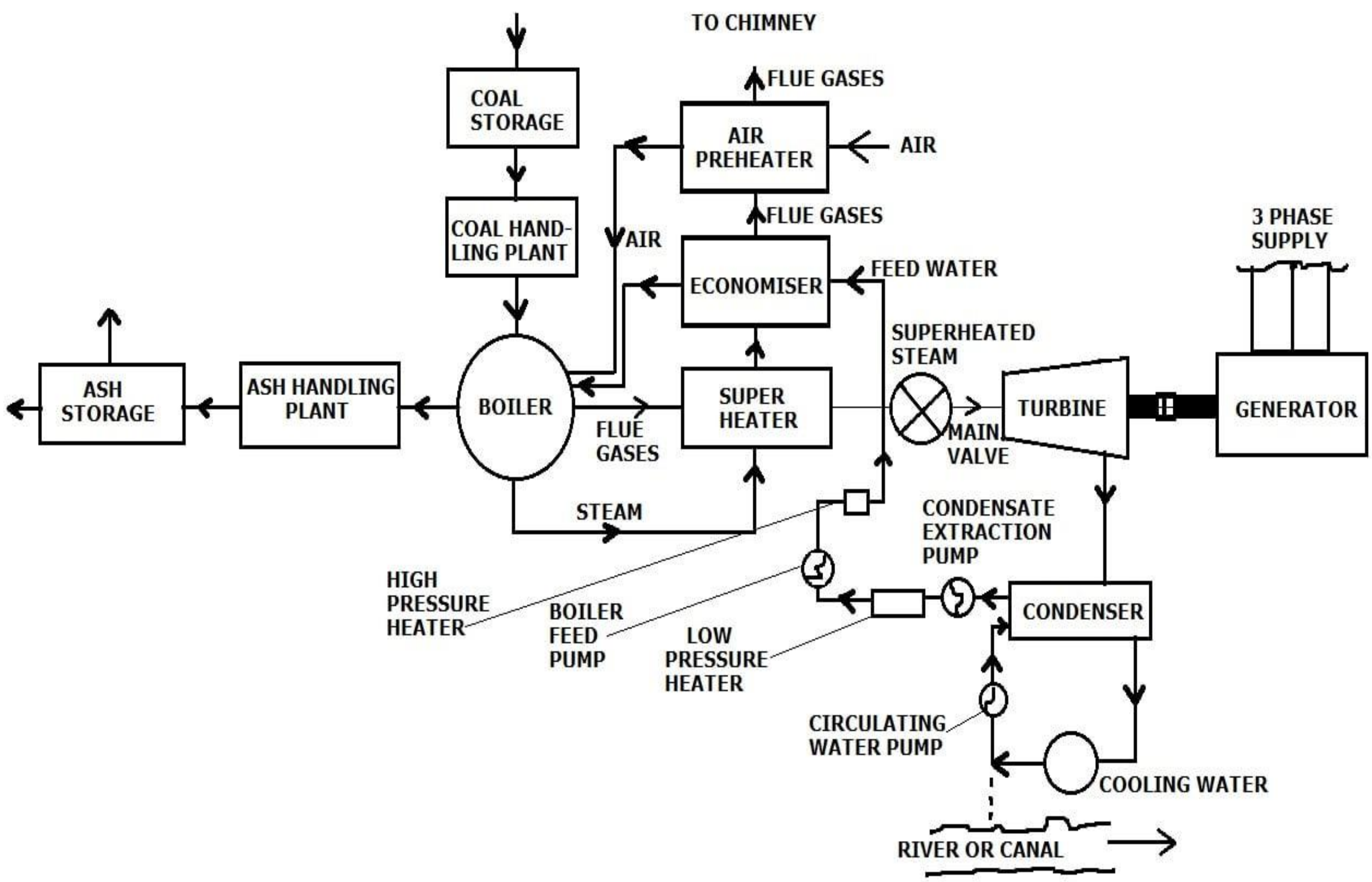


Figure B: Typical layout of Thermal power plant

The major stages in Thermal Power Plant are

1. Coal and Ash Handling Arrangement: The coal and ash handling plant generally consists of:

(i) Coal storage,
(ii) (ii) Coal handling plant,
(iii) (iii) Ash handling plant, and
(iv) (iv) Ash storage.

The coal is stored in a coal storage plant from where it is delivered to the coal handling plant. In coal handling it is crushed into small pieces for proper burning. The coal in the powdered form is burnt in the boiler. The ash produced after burning of coal is removed to ash storage

2. Boiler: The heat of combustion produced by burning coal in the boiler is used to heat up water for producing steam. The steam produced is wet, therefore it is passed through a superheater where it is dried and superheated.

3. Superheater: The wet steam from the boiler is passed through the superheater and superheated by the flue gases on their way to the chimney. This steam is then injected into the steam turbine blades through the main steam value.

4. Economizer: The function of economizer is to preheat the feed water by utilizing the heat of flue gases.

5. Air Preheater: The air is drawn from atmosphere by a forced draught fan and passed through air preheater, where it is heated by flue gases and then admitted to the furnace for the burning of coal.

6. Steam Turbine: The dry and superheated steam from the superheater is fed to the steam turbine to convert heat energy of steam into mechanical energy.

7. Condenser: The used up steam is exhausted to the condenser where it is converted to water which is again fed to the boiler.

8. Alternator: The alternator and turbine are coupled together. The mechanical power is converted to electrical energy by alternator. The output electrical energy is delivered to the bus through transformers, circuit breakers and isolators.

Working:

In these power stations, the heat produced by combustion of coal is utilized to convert water into steam which runs the steam turbine coupled with the alternator to produce electrical energy. After the steam passes through the steam turbine, it is condensed in a condenser and again fed back into the boiler to become steam.

Nuclear power plant:

In Nuclear power plants heat is generated by splitting atoms of radioactive material under controlled conditions. The most amazing feature of a nuclear power plant is that huge amount of electrical energy can be produced from a small amount of nuclear fuel.

The schematic arrangement of a nuclear power station is shown in Figure A.

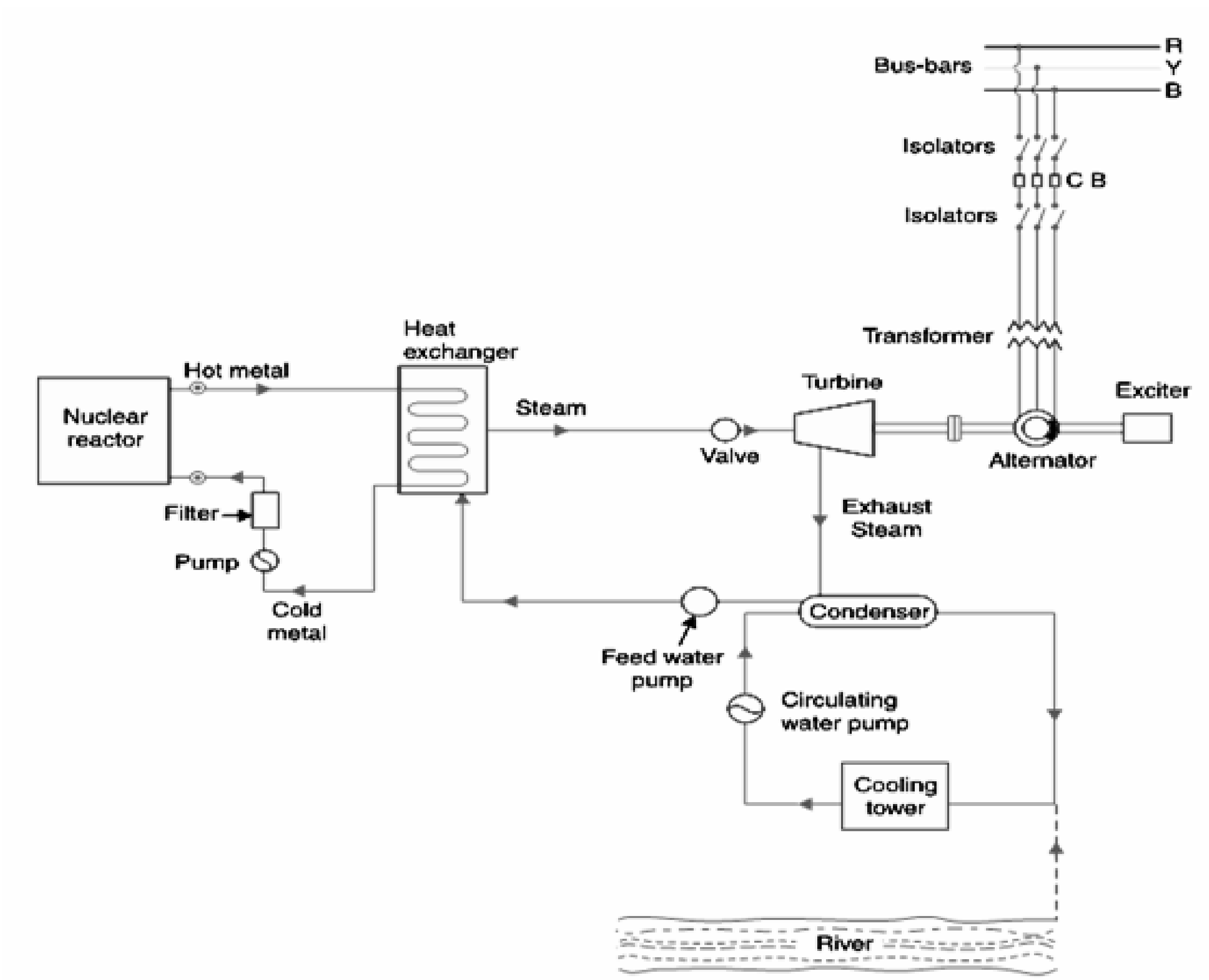


Figure A: Schematic arrangement of a nuclear power plant

The major components of Nuclear Power Plant are

- (i) Nuclear Reactor
- (ii) Coolant
- (iii) Heat Exchanger
- (iv) Steam Turbine
- (v) Condenser
- (vi) Alternator

(i) Nuclear Reactor. It is an apparatus in which nuclear fuel (U235) is subjected to nuclear fission. It controls the chain reaction that starts once the fission is done. If the chain reaction is not controlled, the result will be an explosion due to the fast increase in the energy released. A nuclear reactor is a cylindrical pressure vessel and houses fuel rods of Uranium, moderator and control rods as shown in Figure B.

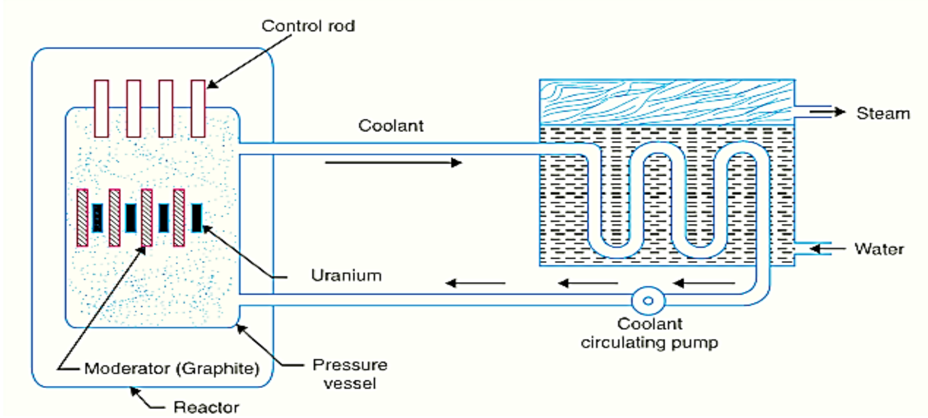


Figure B: Nuclear Reactor

The fuel rods constitute the fission material and release huge amount of energy when bombarded with slow moving neutrons.

The moderator consists of graphite rods which enclose the fuel rods. The moderator slows down the neutrons before they bombard the fuel rods.

The control rods are of cadmium and are inserted into the reactor. Cadmium is strong neutron absorber and thus regulates the supply of neutrons for fission. When the control rods are pushed in deep enough, they absorb most of fission neutrons and hence few are available for chain reaction which, therefore, stops. However, as they are being withdrawn, more and more of these fission neutrons cause fission and hence the intensity of chain reaction (or heat produced) is increased. Therefore, by pulling out the control rods, power of the nuclear reactor is increased, whereas by pushing them in, it is reduced.

(ii) Coolant: The heat produced in the reactor is absorbed by the coolant, generally a sodium metal. The coolant carries the heat to the heat exchanger.

(iii) Heat Exchanger. The coolant gives up heat to the heat exchanger which is utilized in raising the steam. After giving up heat, the coolant is again fed to the reactor.

(iv) Steam Turbine. The steam produced in the heat exchanger is led to the steam turbine through a valve. After doing a useful work in the turbine, the steam is exhausted to condenser.

(v) Condenser: The condenser condenses the steam and converts it to water which is fed to the heat exchanger through feed water pump.

(iv) Alternator: The steam turbine drives the alternator which converts mechanical energy into electrical energy. The output from the alternator is delivered to the bus-bars through transformer, circuit breakers and isolators.

Working :

In Nuclear power plants heat is generated by splitting atoms of radioactive material under controlled conditions. The heat energy thus produced is used in generating steam at high temperature and pressure. This steam drives the steam turbine, which converts steam energy into mechanical energy. The turbine spins the alternator, which converts mechanical energy into electrical energy.

Solar Power Generation:

A solar power system transforms solar radiation directly into electricity. The heart of a solar power system or solar photovoltaic (PV) system are solar cells, which are interconnected to form solar modules (solar panels) and solar arrays. Modules and arrays can be used to charge batteries, operate motors, and to power any number of electrical loads. With the appropriate power conversion equipment, solar power systems can produce alternating current (AC) compatible with any conventional appliances, and can operate in parallel with, and interconnected to the utility. Figure A shows the block diagram of solar energy conversion systems:

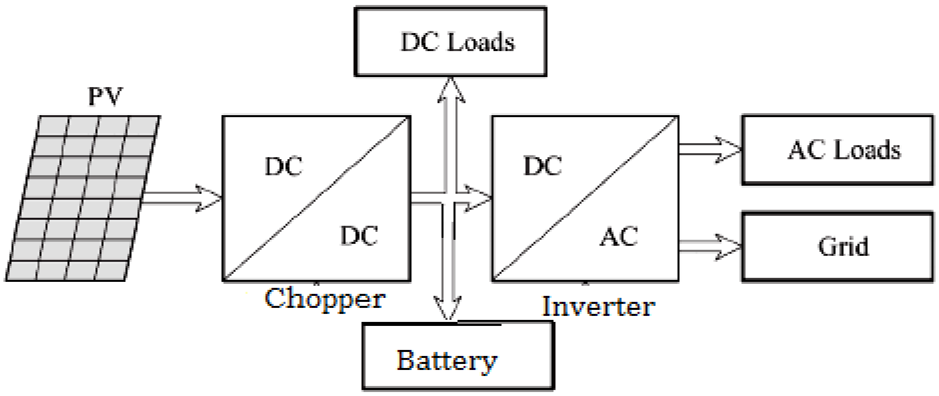


Figure A: Block diagram of solar energy conversion systems

The major components of solar energy conversion system are:

1. Solar panels
2. DC to DC converter
3. DC to AC converter
4. Grid connection equipment

1. Solar Panels: It is the heart of the solar power plant. Solar panels consists a number of solar cells. Solar cells are the energy generating unit, made up of p-type and n-type silicon semiconductor. The energy produced by each solar cell is very small

2. D.C. to D.C. Converter (Choppers): Solar panels produce direct current which is very small and required to be increased so D.C. to D.C. Converter steps up the DC voltage level. DC loads can be fed directly from this supply.

3. Battery: Batteries are used to produce the power back or store the excess energy produced during day, to be supplied during night.

4. D.C. to A.C. Converter (Inverter) Solar panels produce direct current which is required to be converted into alternating current to be supplied to homes or power grid.

Working:

Whenever the sun light falls on the solar panel, the solar cells generate electricity. This DC is stepped up using DC to DC converter. DC loads can be fed directly from this supply. Then DC electricity is fed to an inverter that converts DC to AC which can then be used to feed the AC loads. If a grid connect system is producing more power than is being consumed, the surplus is fed into the mains power grid. When the solar cells are not producing power, for example at night, power is supplied by the main power grid as usual.

Wind Power Generation

Using wind energy is another way of producing electricity. Windmills are devices that harness wind energy, and act like turbines in order to produce electricity. The rotating blades are connected to generators via cables, which transfer kinetic energy to the generators.

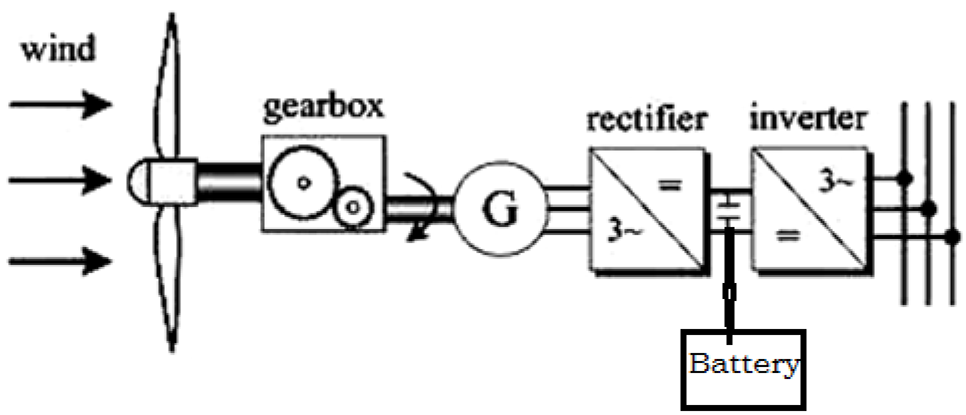
The turbine rotor, gear box and generator are the three main components of wind energy conversion system.

• Rotor converts wind energy to mechanical energy.

• Gear box is used to adapt to the rotor speed to generator speed.

• Generator with the variable speed wind turbine along with electronic inverter absorbs mechanical power and convert to electrical energy.

• The power converter can not only transfer the power from a wind generator, but also improve the stability and safety of the system.



Working:

Wind turbines operate on a simple principle. Wind is merely air in motion. Wind turbines convert kinetic energy from the wind that passes over the rotors into electricity. The kinetic energy in the wind turns two or three propeller-like blades around a rotor. The rotor is connected to the main shaft, which spins a generator to create electricity. Wind turbines are mounted on a tower to capture the most energy. At 100 feet (30 meters) or more above ground, they can take advantage of faster and less turbulent wind. Wind turbines can be used to produce electricity for a single home or building, or they can be connected to an electricity grid for more widespread electricity distribution.

Distributed Generation

Distributed generation is the term used when electricity is generated from sources, often renewable energy sources, near the point of use instead of centralized generation sources from power plants. Hence Distributed generation is also called as on-site generation, Distributed generation reduces the amount of energy lost during transmission because the electricity is generated near the point of consumption. A distributed generation system is designed to employ small-scale power generation technologies to produce electricity close to the end consumers of the power. They meet local peak loads and reduce the size and amount of power lines that need to be built.

In the residential sector, common distributed generation systems include:

- Solar photovoltaic panels
- Small wind turbines
- Natural-gas-fired fuel cells
- Emergency backup generators, usually fueled by gasoline or diesel fuel

In the commercial and industrial sectors, distributed generation can include resources such as:

- Combined heat and power systems
- Solar photovoltaic panels
- Wind
- Hydropower
- Biomass
- Municipal solid waste incineration
- Fuel cells fired by natural gas or biomass

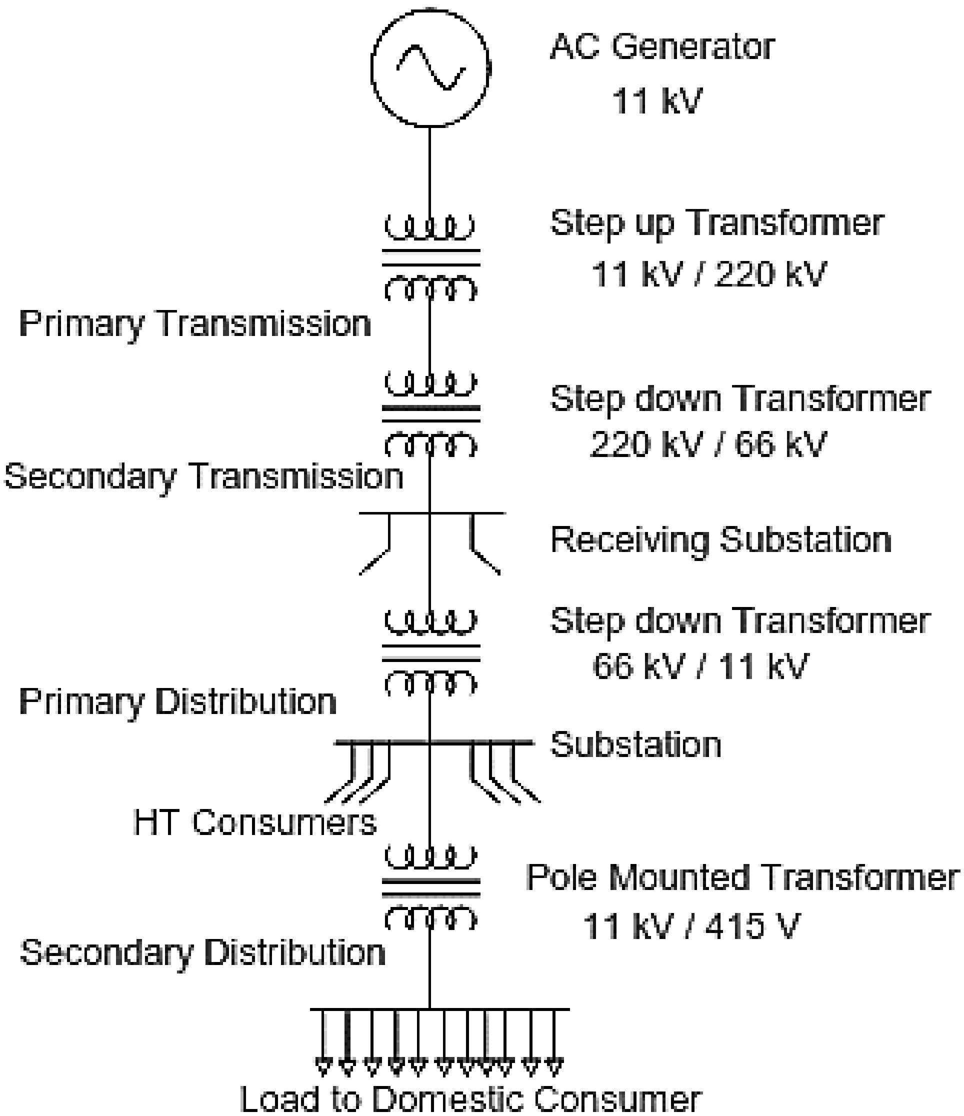
Advantages of Distributed Generation

1. More efficient and less loss involved than centralized generation sources because the generators are closer to the consumers of the energy.
2. Distributed generation can be used to generate electricity at homes and commercial building, using renewable energy sources, such as solar and wind.
3. Distributed generation systems are more reliable than centralized generation systems because multiple small microgrid units are less likely to fail simultaneously than a single large unit.

Disadvantages of Distributed Generation

1. Distributed generation single units take up space and are located closer to the consumers, so they may cause land-use concerns and be displeasing to the eye.
2. Microgrids have many regulations that can be difficult to comply with. For example, cybersecurity is a major regulatory issue that a microgrid owner must be aware of when collecting personal information from individuals.
3. Distributed generation systems that involve burning fossil fuels can produce the same types of impacts as larger fossil-fuel-powered plants on a smaller scale, but closer to a populated area.

Single Line Diagram of AC Power Transmission and Distribution System



Generating station

- The electrical power is generated in the thermal / hydro / nuclear power station.
- The generation voltage is 11 kV .
- The generated power is transmitted to different areas which are far away from generating station.
- The generated voltage is stepped up for transmission due to following reasons. As the generating voltage increases

  (1) Conductor volume (weight) decreases

  (2) Efficiency of transmission line increases

  (3) Percentage voltage drop decreases.

Primary transmission

- The step up transformers steps up the generating voltage from 11 kV to 132 kV / 220 kV.
- It is transmitted by three phase, three wire supply system.

Secondary transmission

- The primary transmission line terminates at the receiving substation.
- The transmission line voltage steps down from 132 kV / 220 kV to 66 kV at the receiving end substation or receiving stations.
- The electrical power is transmitted at voltage level of 66 kV to different substation.
- This is called as secondary transmission.

Primary distribution

- The secondary transmission line terminates at the substation where the voltage is reduced from 66 kV / 11 kV by step down transformer.
- The 11 kV voltage is transmitted to different area of city by transmission line.
- The primary transmission is done through 3 – phase, 3 – wire system which supply power to HT consumers.

Secondary distribution

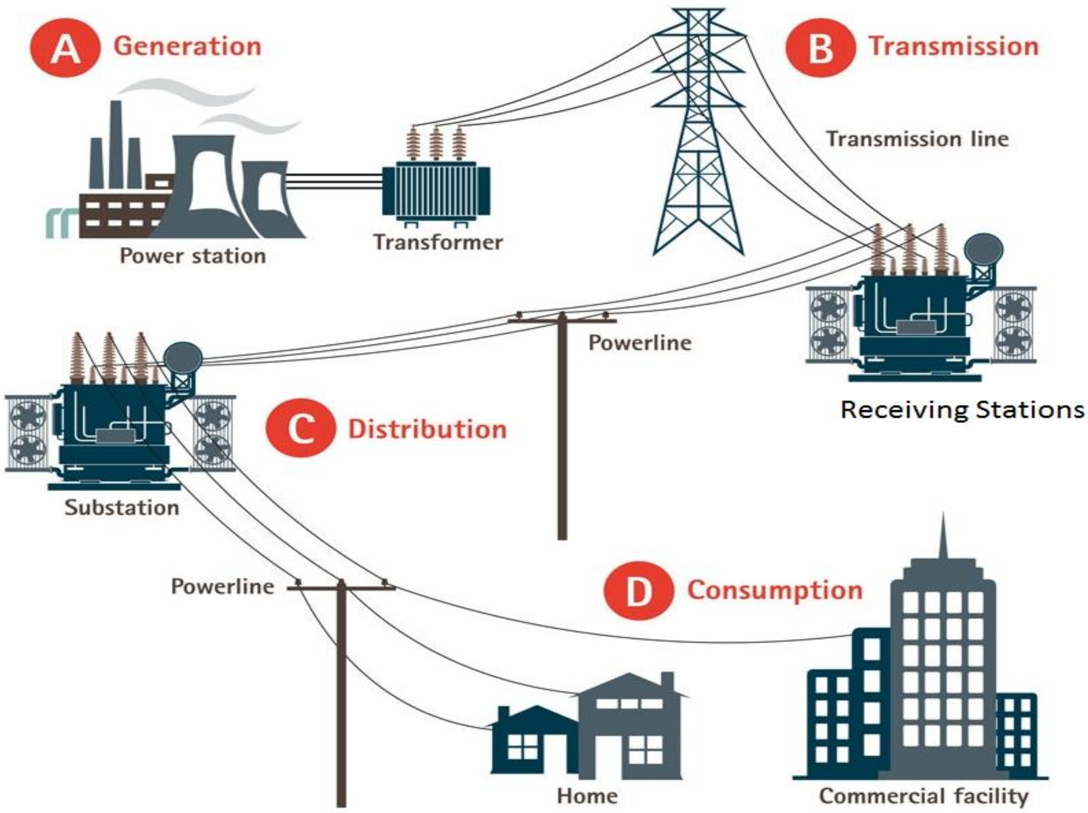
- The electrical power from primary distribution line is delivered to pole mounted distribution transformer.
- The pole mounted delta / star transformer steps down the 11 kV to 440 V.
- The residential consumer gets single phase supply through one phase and neutral (Phase R and N, Phase Y and N, Phase B and N).

Electrical Grid or Power Grid

Electrical grid or power grid is defined as the network which interconnects the generation, transmission and distribution unit. It supplies the electrical power from generating unit to the distribution unit.

(or)

An electrical grid or power grid is an interconnected network for delivering electricity from producers to consumers.



The Indian Power system for planning and operational purposes is divided into five regional grids. The integration of regional grids, and there by establishment of National Grid was conceptualized in early nineties

- Initially, State grids were inter-connected to form regional grid and India was demarcated into 5 regions namely Northern, Eastern, Western, North Eastern and Southern region.
- In October 1991 North Eastern and Eastern grids were connected.
- In March 2003 WR and ER-NER were interconnected.
- August 2006 North and East grids were interconnected thereby 4 regional grids Northern, Eastern, Western and North Eastern grids are synchronously connected forming central grid operating at one frequency.
- On 31st December 2013, Southern Region was connected to Central Grid in Synchronous mode with the commissioning of 765kV Raichur-Solapur Transmission line thereby achieving 'ONE NATION'-'ONE GRID'-'ONE FREQUENCY'.

Need for Interconnection of Grids

1. The interconnection of the grid provides the best use of power resource and ensures great security to supply.
2. It makes the system economical and reliable.
3. The generating stations are interconnected for reducing the reserve generation capacity in each area.
4. If there is a sudden increase in load or loss of generation in a zone, then it borrows from the adjacent interconnected area.

Conditions for Grid Connection

1. The phase sequence *(*or phase rotation*)* of the three phases of the generator must be the same as the phase sequence of the three phases of the grid.
2. The magnitude of the sinusoidal voltage produced by the generator must be equal to the magnitude of the sinusoidal voltage of the grid.
3. The frequency of the sinusoidal voltage produced by the generator must be equal to the frequency of the grid.
4. Phase angle between difference the voltage produced by the generator and the voltage produced by the grid must be zero.

Smart Grid

Smart Grid is an Electrical Grid with Automation, Communication and IT systems that can monitor power flows from points of generation to points of consumption (even down to appliances level) and control the power flow or curtail the load to match generation in real time or near real time.

Smart Grids can be achieved by implementing efficient transmission & distribution systems, system operations, consumer integration and renewable integration.

Benefits of Smart Grid:

The benefits associated with the Smart Grid include:

- More efficient transmission of electricity
- Quicker restoration of electricity after power disturbances
- Reduced operations and management costs for utilities, and ultimately lower power costs for consumers
- Reduced peak demand, which will also help lower electricity rates
- Increased integration of large-scale renewable energy systems
- Better integration of customer-owner power generation systems, including renewable energy systems
- Improved security.

Regulatory Bodies

1. Central Electricity Regulatory Commission (CERC)

The CERC and SERCs were set up by the 1998 Act (later reconstituted under the Electricity Act, 2003) with the objectives of depoliticizing the sector and incentivising private investment. The Central Electricity Regulatory Commission (CERC) intends to promote competition, efficiency and economy in bulk power markets, improve the quality of supply, promote investments and advise government on the removal of barriers to bridge the demand supply gap and thus foster the interests of consumers. The Central Electricity Regulatory Commission (CERC) regulates tariff for generating companies owned or controlled by the Central Government and those with an inter-State dimension, and for inter-State transmission of electricity.

As entrusted by the Electricity Act, 2003 the Commission has the responsibility to discharge the following functions:

- To regulate the tariff of generating companies owned or controlled by the Central Government.
- To regulate the tariff of generating companies other than those owned or controlled by the Central Government, if such generating companies enter into or otherwise have a composite scheme for generation and sale of electricity in more than one State.
- To regulate the inter-State transmission of electricity.
- To determine tariff for inter-State transmission of electricity.
- To issue licenses to persons to function as transmission licensee and electricity trader with respect to their inter-State operations.
- Improve access to information for all stakeholders.
- To adjudicate upon disputes involving generating companies or transmission licensee
- To specify Grid Code having regard to Grid Standards.
- To specify and enforce the standards with respect to quality, continuity and reliability of service by licensees.
- To fix the trading margin in the inter-State trading of electricity, if considered, necessary.

CERC can advise the government on National Electricity Policy and tariff policy, and on promoting efficiency, competition and private investment in the sector. It is bound by Central Government/ Central Electricity Autority (CEA) policy in the discharge of its functions.

2. State Electricity Regulatory Commission (SERC)

SERCs are appointed by the State Governments to regulate generation, transmission, wheeling and distribution of electricity within the State. The main function of SERC are.

- They regulate power purchase and procurement processes, issue licenses and determine tariffs for electricity operations in the State.
- They are also responsible for adjudicating disputes between licensees and distribution companies.
- They have similar standards setting and policy advisory roles as the CERC in relation to the State, and are bound to follow national electricity and tariff policy in the discharge of their functions.

Electricity Rates or Tariff

Today's interconnected power systems supply a number of consumers. The supply companies have to sell their electricity at such a rate that it covers the costs of generation, transmission, distribution, the salaries of the employees, the interest and depreciation and the profit targeted by the company. This **rate at which electrical energy is sold to the consumers is termed as 'tariff.'**

The cost of generation of electricity will depend upon various factors such as Connected Load, Maximum Demand, Load factor, Demand Factor, Diversity Factor, Plant Capacity Factor and Use Factor. These, in turn, will depend upon the type of load and load conditions. Hence, the tariff is different for different type of loads (and hence different consumers). Therefore, while fixing the tariff, we have to consider various consumers (industrial, domestic, commercial, etc.) and their requirements.

Characteristics of Tariff

A tariff must have following desirable characteristics −

- **Simplicity** – The tariff should be simple so that an ordinary consumer can easily understand it.
- **Fairness** – The tariff should be fair so that the different types of consumers are satisfied with the rate of charge of electricity.
- **Attractive** – The tariff should be attractive so that large number of consumers are encouraged to use electricity.
- **Proper Return** – The tariff should be such that it ensures the proper return from each consumer, i.e. the total receipt from the consumers must be equal to the cost of production and distribution with reasonable profit.

Keeping in mind the above factors, various types of tariff have been designed. The most commonly used are given below.

Various Types of Electricity Tariff

1. Simple Tariff

In this type of tariff, a fixed rate is applied for each unit of the energy consumed. It is also known as a uniform tariff. The **rate per unit of energy does not depend upon the quantity** of energy used by a consumer.

The price per unit (1 kWh) of energy is constant. This energy consumed by the consumer is recorded by the energy meters. Graphically, it can be represented as follows:

Figure: Simple Tariff

Advantages:

- Simplest method
- Easily understandable and easy to apply
- Each consumer has to pay according to his utilization

Disadvantages

- There is no discrimination according to the different types of consumers.
- The cost per unit is high.
- There are no incentives
- If a consumer does not consume any energy in a particular month, the supplier cannot charge any money even though the connection provided to the consumer has its own costs.

Application

- Generally applied to tube wells used for irrigation purposes.

2. Flat Rate Tariff

In this tariff, different types of consumers are charged at different rates of cost per unit (1 kWh) of electrical energy consumed. Different consumers are grouped under different categories. Then, each category is charged at a fixed rate similar to Simple Tariff. The different rates are decided according to the consumers, their loads and load factors.

Graphically, it can be represented as follows:

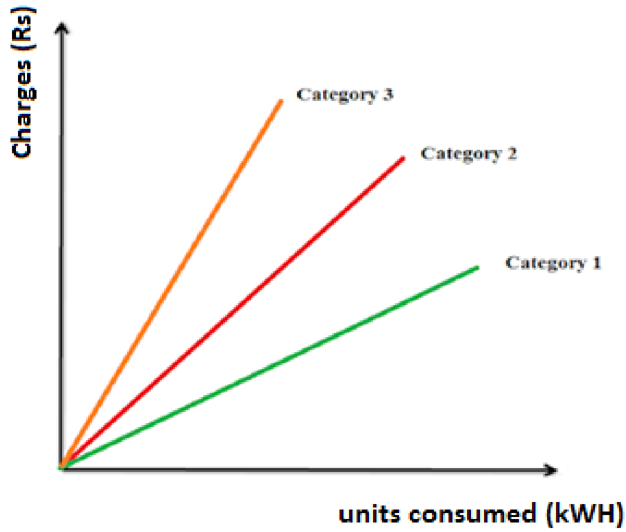


Figure: Flat rate Tariff

Advantages

- More fair to different consumers.
- Simple calculations.

Disadvantages

- A particular consumer is charged at a particular rate. But there are no incentives for the consumer.
- Since different rates are decided according to different loads, separate meters need to be installed for different loads such as light loads, power loads, etc. This makes the whole arrangement complicated and expensive.
- All the consumers in a particular “category” are charged at the same rates. However, it is fairer if the consumers that utilize more energy be charged at lower fixed rates.

Application

- Generally applied to domestic consumers.

3. Block Rate Tariff

In this tariff, the first block of the energy consumed (consisting of a fixed number of units) is charged at a given rate and the succeeding blocks of energy (each with a predetermined number of units) are charged at progressively reduced rates. The rate per unit in each block is fixed.

For example, the first 50 units (1st block) may be charged at 3 rupees per unit; the next 30 units (2nd block) at 2.50 rupees per unit and the next 30 units (3rd block) at 2 rupees per unit.

Graphically, it can be represented as follows:

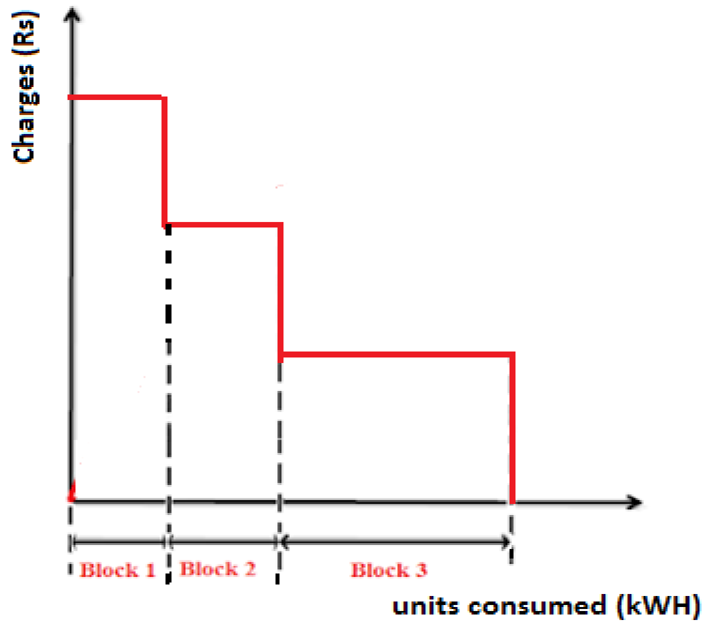


Figure: Block Rate Tariff

Advantages

- Only 1 energy meter is required.
- Incentives are provided for the consumers due to reduced rates. Hence consumers use more energy. This improves load factor and reduces cost of generation.

Disadvantages

- If a consumer does not consume any energy in a particular month, the supplier does not charge any money even though the connection provided to the consumer has its own costs.

Application

- Generally applied to residential and small commercial consumers.

4. Two Part Tariff

In this tariff scheme, the total costs charged to the consumers consist of two components: Fixed charges and Running charges. It can be expressed as:

Total Cost = [A (kW) + B (kWh)] Rs.

Where, A = charge per kW of max demand (i.e. A is a constant which when multiplied with max demand (kW) gives the total fixed costs.)

B = charge per kWh of energy consumed (i.e. B is a constant which when multiplied with units consumed (kWh), gives total running charges.)

The fixed charges will depend upon maximum demand of the consumer and the Running charge will depend upon the energy (units) consumed.

Advantages

- If a consumer does not consume any energy in a particular month, the supplier will get the return equal to the fixed charges.

Disadvantages

- Even if a consumer does not use any electricity, he has to pay the fixed charges regularly.
- The maximum demand of the consumer is not determined. Hence, there is error of assessment of max demand and hence conflict between the supplier and the consumer.

Application

- Generally applied to industrial consumers with appreciable max demand.

5. Maximum Demand Tariff

In this tariff, the energy consumed is charged on the basis of maximum demand. The units (energy) consumed by consumer is called maximum demand. The maximum demand is calculated by a maximum demand meter. This removes any conflict between the supplier and the consumer as it were the two part tariff. It is similar to two-part tariff.

Application

- Generally applied to large industrial consumers.

6. Power Factor Tariff

In this tariff scheme, the power factor of the consumer's load is also considered. We know that power factor is an important parameter in power system. For optimal operation, the pf must be high. Low pf will cause more losses and imbalance on the system. Hence the consumers who have low pf loads will be charged more. It can be further divided into the following types:

(I) KVA Maximum Demand Tariff

In this type of tariff, the fixed charges are made on the basis of maximum demand in kVA instead of KW.

We know that power factor = kW / kVA

Hence, the pf is inversely proportional to kVA demand. Hence, a consumer having low power factor load will have to pay more fixed charges. This gives the incentive to the consumers to operate their load at high power factor. Generally, the suppliers ask the consumers to install power factor correction equipment.

(II) KW and KVAR Tariff

In this tariff scheme, the active power (kW) consumption and the reactive power (kVAR) consumption is measured separately. A consumer having low power factor load will have to pay more fixed charges.

(III) Sliding Scale Tariff

In this type of tariff scheme, an average power factor (generally 0.8 lagging) is taken as reference. Now, if the power factor of the consumer's loads is lower than the reference, he is penalized accordingly. Hence, a consumer having low power factor load will have to pay more fixed charges. Also, if the pf of the consumer's load is greater than the reference, he is awarded with a discount. This gives incentives to the consumers. It is usually applied to large industrial consumers.

7. Three Part Tariff

In this scheme, the total costs are divided into 3 sections: Fixed costs, semi-fixed costs and running costs.

Total Charges = [A + B (kW) + C (kWh)]

Where, A = fixed charges,

B = charge per kW of max demand (i.e. B is a constant which when multiplied with max demand (kW) gives the total fixed costs.)

C = charge per kWh of energy consumed (i.e. C is a constant which when multiplied with units consumed (kWh), gives total running charges.)

Application

- This type of tariff is generally applied to big consumers.